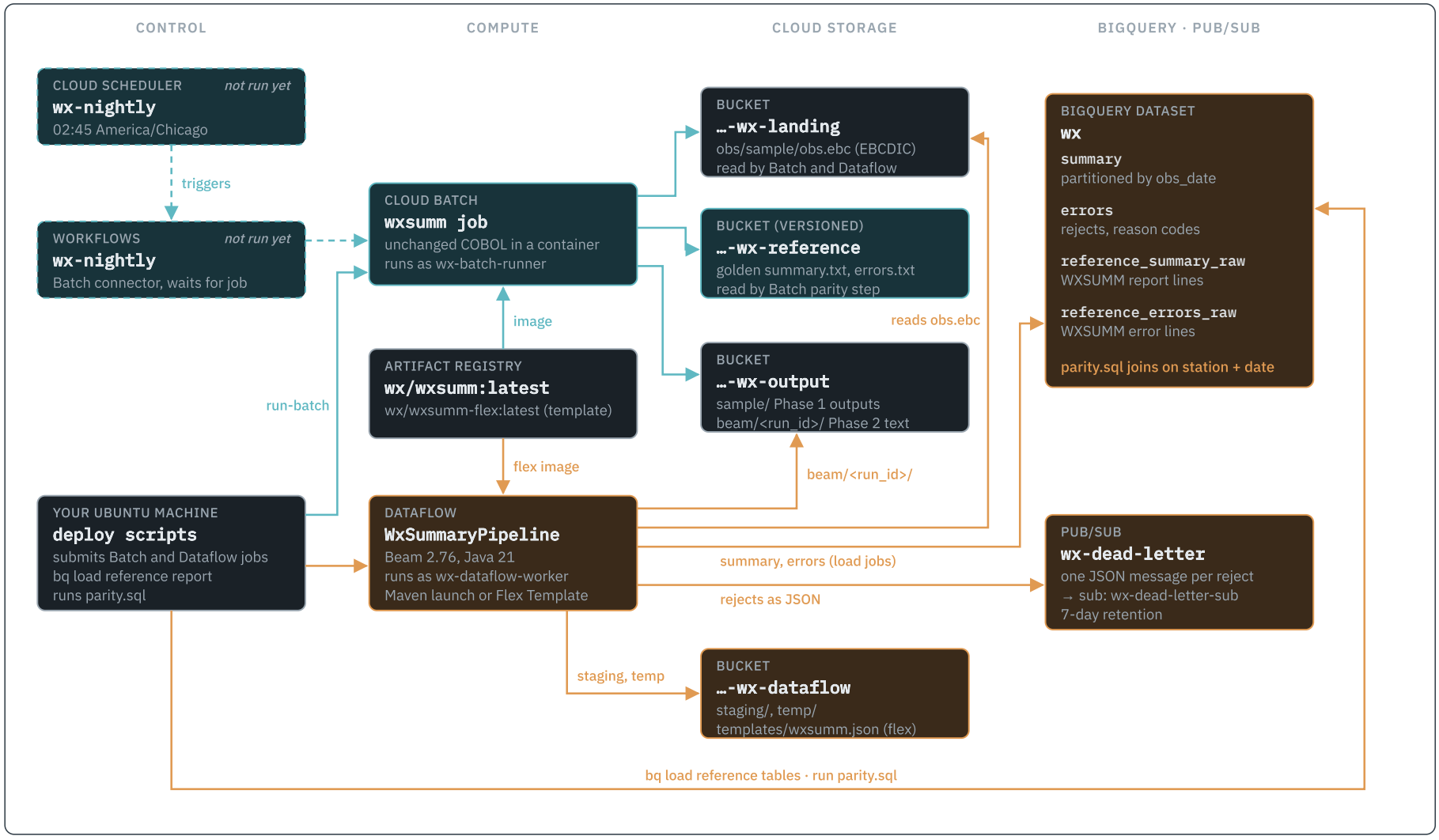
WXSUMM on GCP

`project wx-modernization-0 · us-central1 · as of 14 Oct 2026`

A COBOL weather batch job (WXSUMM) moved to Google Cloud in two phases. Phase 1 runs the unchanged program in a container on Cloud Batch. Phase 2 re-expresses the same logic as a Java Beam pipeline on Dataflow, writing to BigQuery and Pub/Sub. Both phases read the same EBCDIC input and are checked against the same golden output.

Phase 1: lift and shift — Phase 2: refactor — Defined in the repo, not run yet



Resources

RESOURCE	SERVICE	PHASE	ROLE	STATUS
wxsumm job	Cloud Batch	1	Runs unchanged COBOL in a container, then a byte-for-byte parity check	Ran, parity OK
wx-nightly	Workflows	1	Runs the Batch job and fails on any non-success state	Not deployed
wx-nightly	Cloud Scheduler	1	Starts the workflow daily at 02:45 Chicago time	Not deployed
WxSummaryPipeline	Dataflow	2	Decode, validate, summarize per station-day; detects keys the legacy control break would split. Launched directly from Maven and from the Flex Template	Ran both ways, parity OK
wx.summary	BigQuery	2	One row per station-day per run; NUMERIC values; partitioned by obs_date	Loaded
wx.errors	BigQuery	2	Rejected records with legacy reason codes R01–R05 (R06 = undecodable)	Loaded
wx.reference_*_raw	BigQuery	2	WXSUMM report, one line per row; parsed by column position in parity.sql	Loaded
wx-dead-letter	Pub/Sub	2	One JSON message per reject; subscription wx-dead-letter-sub	Not checked yet
…-wx-landing	Cloud Storage	1 + 2	EBCDIC input, obs/sample/obs.ebc	In use
…-wx-reference	Cloud Storage	1	Golden outputs from the COBOL run (versioned)	In use
…-wx-output	Cloud Storage	1 + 2	Phase 1 report files; Phase 2 text output per run	In use
…-wx-dataflow	Cloud Storage	2	Dataflow staging and temp; Flex Template spec	In use
wx/wxsumm	Artifact Registry	1	Container image: GnuCOBOL runtime and WXSUMM	Pushed
wx/wxsumm-flex	Artifact Registry	2	Flex Template launcher image, built by Cloud Build	Built, launched
wx-batch-runner	Service account	1	Batch VM identity: read landing and reference, write output	Active
wx-workflow	Service	1	Creates Batch jobs as wx-batch-runner	Created

	account			
`wx-dataflow-worker`	Service account	2	Dataflow workers: buckets, BigQuery load jobs, publish to the topic	Active

Not shown

- Cloud Logging receives logs from both the Batch job and the Dataflow job.
- Cloud Build builds the Flex Template image when `deploy_phase2.sh flex-build` runs.
- BigQuery parity: the Dataflow output matches the WXSUMM reference row for row (20 summaries, 5 rejects), checked by `parity.sql`.